

Algebra 1
Unit 1
Lesson 6 Practice

Name **Answer Key** _____
Date _____
Period _____

1. Explain mathematically why the expressions $\left(a^{\frac{1}{c}}\right)^b$ is equivalent to $(a^b)^{\frac{1}{c}}$ using the expressions $\left(4^{\frac{1}{2}}\right)^5$ and $(4^5)^{\frac{1}{2}}$.

$$\left(4^{\frac{1}{2}}\right)^5 = (4)^{5 \cdot \frac{1}{2}} = (4)^{\frac{5}{2}} = (4^5)^{\frac{1}{2}}$$

2. Identify all of the expressions equivalent to $\left(49^{\frac{1}{2}}\right)^5$.

- $49^{\frac{2}{5}}$
 - $(49^5)^{\frac{1}{2}}$
 - $\sqrt[5]{49^2}$
 - $49^{\frac{5}{2}}$
 - $(\sqrt{49})^5$
- $$\begin{aligned}\left(49^{\frac{1}{2}}\right)^5 &= 49^{\frac{5}{2}} \\ &= \sqrt[2]{49^5} \\ &= \sqrt{49^5}\end{aligned}$$

For questions 3-7, complete each row with equivalent expressions.

	$a^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
3.	$256^{\frac{3}{4}}$	$\left(256^{\frac{1}{4}}\right)^3$	$(256^3)^{\frac{1}{4}}$	$(\sqrt[4]{256})^3$	$\sqrt[4]{256^3}$
4.	$64^{\frac{3}{2}}$	$\left(64^{\frac{1}{2}}\right)^3$	$(64^3)^{\frac{1}{2}}$	$(\sqrt{64})^3$	$\sqrt{64^3}$
5.	$128^{\frac{5}{7}}$	$\left(128^{\frac{1}{7}}\right)^5$	$(128^5)^{\frac{1}{7}}$	$(\sqrt[7]{128})^5$	$\sqrt[7]{128^5}$
6.	$(-27)^{\frac{2}{3}}$	$\left(-27^{\frac{1}{3}}\right)^2$	$(-27^2)^{\frac{1}{3}}$	$(\sqrt[3]{-27})^2$	$\sqrt[3]{-27^2}$
7.	$81^{\frac{3}{2}}$	$\left(81^{\frac{1}{2}}\right)^3$	$(81^3)^{\frac{1}{2}}$	$(\sqrt{81})^3$	$\sqrt{81^3}$

For questions 8-10, rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$.

8. $\sqrt[3]{(-44)^5}$

$a = (-44)$

$b = 5$

$c = 3$

$= (-44)^{\frac{5}{3}}$

9. $(\sqrt{289})^3$

$a = 289$

$b = 3$

$c = 2$

$289^{\frac{3}{2}}$

10. $\frac{1}{\sqrt[5]{30^2}}$

$a = 30$

$b = 2$

$c = 5$

$30^{\frac{2}{5}} \rightarrow \frac{1}{30^{\frac{2}{5}}} \text{ or } 30^{-\frac{2}{5}}$